

The weak data assumption specifies that β , while arbitrary, is bounded away from zero. Just as the weak learning assumption might not hold in classical boosting, the weak data assumption might not hold in our setting. We argue that iteratively filtering the weak labeler’s output should result in a precipitous drop in the fraction of correctly-labeled examples. As an example, suppose human labelers provide good responses to the β_1 easiest coding prompts in some prompt set. One should expect that asking similarly-qualified labelers to respond to the remaining prompts results in a $\beta_2 \ll \beta_1$ yield of quality responses, as all but the easiest prompts have been answered. In contrast, a continually improved LLM endows a human with more flexibility for future responses, such as rewriting nearly high-quality solutions provided by the last iteration of LLM, making a non-vanishing β a much more reasonable assumption.

While it keeps the setting simple to presume that q can be evaluated on any labeled example, this is an overly permissive assumption. LLM-generated synthetic data can be made to include reasoning traces, and often produces responses that the LLM itself can verify as high quality. This facilitates the construction of automated quality checkers, which are much more difficult to construct when the labels are produced by a human, and therefore contain reasoning traces and responses that are unfamiliar to the current generation of LLM. This is born out in the literature, where quality verification of LLM-generated synthetic data is relatively easy to implement [Singh et al., 2024, Yang et al., 2024, Zelikman et al., 2022].

Finally, and somewhat remarkably, Algorithm 2 with $\alpha = \epsilon$ achieves the same finite-time error rate as this baseline while only ever evaluating the quality of LLM-generated data. Thus, the approach taken in practice matches the convergence rate that would be experienced under a much more powerful set of assumptions.

F Additional Experimental Results

In this section, we provide additional experimental results. Specifically, we conduct the following experiments:

- **Off-policy boosting:** We fixed the learner to be Gemma 2 2B and switched out the labeler from Gemma 2 2B (on-policy) to either Gemma 1 2B or Gemma 7B. The stronger off-policy Gemma 7B model outperforms the on-policy Gemma 2 2B model. Results are pictured in Figure 5. The weaker Gemma 1 2B significantly underperforms on-policy Gemma 2 2B, but after 5 rounds roughly approaches the performance of *filter only*. Note that the Gemma 1 2B model achieves only 11% on GSM8K, and based on our theory, we should expect to require more iterations with a weaker labeler.
- **Running longer and observing improvement plateau:** In several prior works, it was reported that improvement plateaus after a few rounds. We run some experiments longer (>5 rounds). Indeed, in Figure 6 we see limited improvements in test accuracy ($\approx 2\%$) from additional rounds, despite persistent improvements in training accuracy.

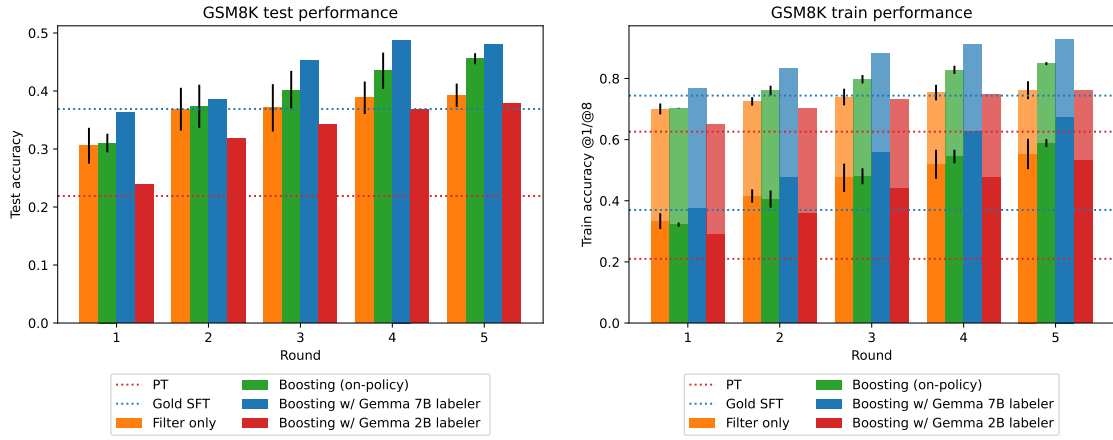


Figure 5: We experiment with *off-policy labelers* on GSM8K, plotting performance across rounds. *Boosting (on-policy)* is the setting in all prior experiments, employing Gemma 2 2B PT as the labeler. We see improvement from using the stronger Gemma 7B as our labeler. The weaker Gemma 1 2B performs much worse, but approaches the results from *Filter only* after 5 rounds.

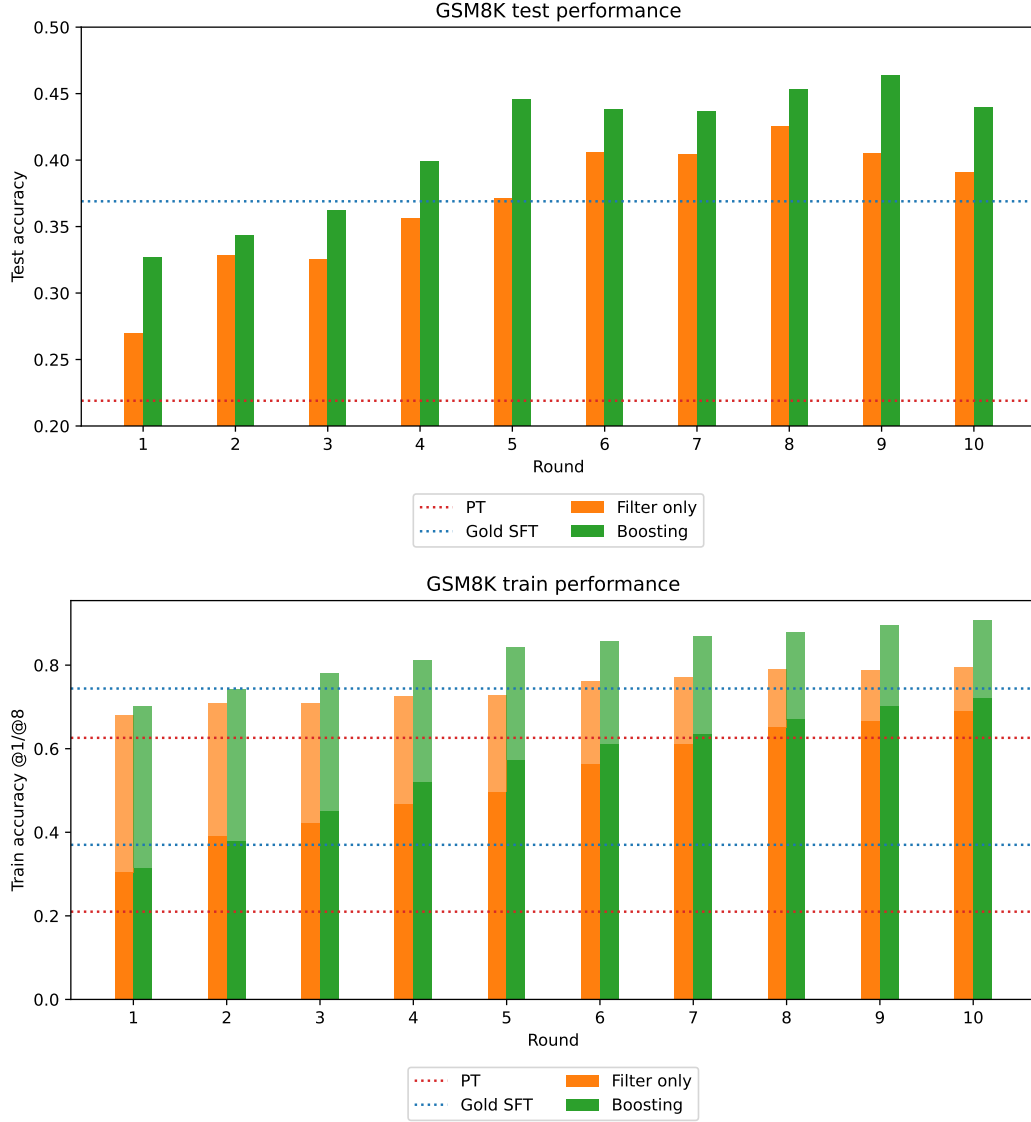


Figure 6: We plot test and train performance of *Filter only* and *Boosting* on GSM8K across more rounds (10). We see limited improvements in test accuracy ($\approx 2\%$) despite persistent improvements in train accuracy.